

on the same hard disk. You have to plug in another storage device on which data can be stored.

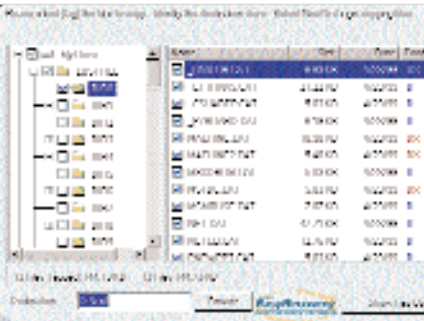
STEP 3 Next, make a bootable rescue floppy disk from the recovery software. Insert this floppy disk and boot your machine in the MS-DOS mode. Do not, by any chance, boot your machine from the hard disk. In order to boot your machine, make sure you go and change the boot sequence from the BIOS.

STEP 4 After you've booted in DOS mode, insert the recovery software floppy diskette. One of the most widely used and effective software in this category is Easy Recovery from Ontrack. You can also use other software such as Stellar Phoenix Data Recovery Software, Final Data, etc.

STEP 5 After inserting the recovery software, you will get an option where you will have to select the primary hard disk (that is, the one that has crashed) as the 'recovery source' and the second hard disk as the 'restore destination'.

STEP 6 The software then analyses the data that it has to recover from the crashed hard disk. In order to recover the data, the software tries to read the two copies of the file allocation table (FAT) and then reads the raw data on the hard drive, trying to match it to the FAT.

The data that does not match the FAT is generally the data that you are trying to recover. Once it finds raw data on the disk



Select the files and destination path of the files to be recovered

that equals a file but no similar entry in the FAT, it marks the data as a recoverable file and adds it to the existing list of files it has from the FAT. All the files that exist on the hard drive, whether in the FAT or detected by the software, are shown as recoverable files. Once this is done, you can choose what you want to recover, and

glossary

Tracks and Sectors: A hard disk consists of platters on which data is stored. These platters consist of tens of thousands of tracks that are tightly integrated in concentric circles. These tracks contain loads of vital information and thus are broken down into sectors. Hence, a sector is the smallest addressable unit that can hold 512 bytes of information.

SMART: Self-Monitoring Analysis and Reporting Tool (SMART) system uses internal hard disk monitoring technology to analyse your hard disk and prevent hard disk failure or data loss.

Bad Sector: A sector that cannot be used due to a physical flaw on the disk is called a bad sector.

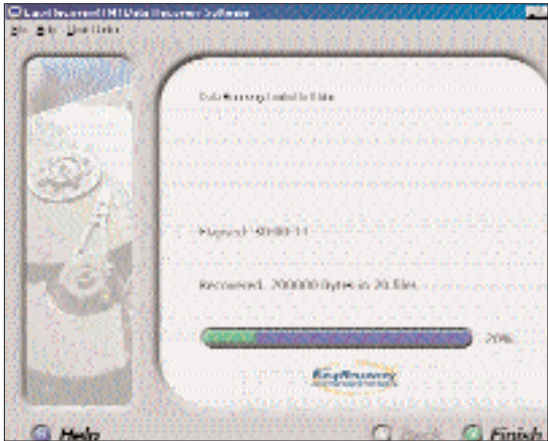
Clusters: A cluster is a logical unit of storage on a hard disk or a floppy drive. The

size of a cluster varies from 512 bytes to 256 kilobytes depending on the size of the partition and the file system.

FAT: The FAT (file allocation table) tells the Disk Operating System which portions of the disk belong to each file. The FAT links together all the clusters belonging to each file, no matter where they are on the disk.

Partition: Whenever a hard disk is formatted, it will have one partition by default. Hence, a partition is a formatted section of a hard disk. Large hard disks can be formatted to multiple partitions and each partition acts like an individual or a separate hard drive.

Boot disk: A special floppy boot disk allows your computer to boot even if it cannot boot from the hard disk. This disk is also known as a System Rescue Disk.



You can recover data using utilities like Easy Recovery

it will be copied to the destination drive. The whole process might take hours to complete depending on the data.

Warranty woes

Like all products, even hard disks are

covered under specific warranty periods, depending on the manufacturer. Usually, a hard disk (IDE) carries a one- to three-year warranty from the date of purchase, and certain SCSI hard disks carry warranties up to five years. If during this period your hard disk crashes or malfunctions, you can return it to the dealer so that it can be sent for further analysis to the company's RMA centre. There the hard disk will be repaired and if successful, the same hard disk will be returned to you. If unsuccessful, you will be given a hard disk with a similar model that has also come in for repairs.

In case your hard disk is not repairable and a similar model is not available for replacement, you can either opt for

Are High-speed Hard Disks More Vulnerable to Crashes?

Hard disks with a rotational speed of 5,400 rpm use metal substrate disks, whereas the newer hard disks that run at 7,200 rpm or even 10,000 rpm use glass substrate disks. Using glass substrate disks has its own advantages, such as better ability to withstand shock, and reduction in surface defects in turn helps reduce read and write errors.

However, these could die out soon as glass substrate disks are more brittle than

metal ones and more over, the magnetic film coating applied on the glass substrate disk peels off over time, thereby damaging the hard disk.

Another important factor is heat. The heat generated by hard disks with a rotational speed of 7,200 rpm and above is immense. This sometimes causes the platter dimensions to change, which is especially dangerous when data is read or written on a hard drive.